

## Visualization Analysis & Design

### Arrange Tables (Ch 7) I

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### Focus on Tables

### Spatial

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### Keys and values

- key
  - independent attribute
  - used as unique index to look up items
  - simple tables: 1 key
  - multidimensional tables: multiple keys

- value
  - dependent attribute, value of cell

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### Keys and values

- key
  - independent attribute
  - used as unique index to look up items
  - simple tables: 1 key
  - multidimensional tables: multiple keys

- value
  - dependent attribute, value of cell
- classify arrangements by keys used
  - 0, 1, 2, ...

0

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Idiom: scatterplot

- express values (magnitudes)
- quantitative attributes
- no keys, only values

[A layered grammar of graphics. Wickham. Journ. Computational and Graphical Statistics 19:1 (2010), 3–28.]

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Idiom: scatterplot

- express values (magnitudes)
- quantitative attributes
- no keys, only values
- data
- 2 quant attribs
- mark: points
- channels
- horiz + vert position

[A layered grammar of graphics. Wickham. Journ. Computational and Graphical Statistics 19:1 (2010), 3–28.]

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Idiom: scatterplot

- express values (magnitudes)
- quantitative attributes
- no keys, only values
- data
- 2 quant attribs
- mark: points
- channels
- horiz + vert position
- tasks
- find trends, outliers, distribution, correlation, clusters
- scalability
- hundreds of items

[A layered grammar of graphics. Wickham. Journ. Computational and Graphical Statistics 19:1 (2010), 3–28.]

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Scatterplots: Encoding more channels

- additional channels viable since using point marks
  - color
  - size (1 quant attribute, used to control 2D area)
- note radius would mislead, take square root since area grows quadratically
  - shape

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Scatterplot tasks

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Scatterplot tasks

- correlation

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Scatterplot tasks

- correlation
- clusters/groups, and clusters vs classes

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Some keys

0

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Some keys: Categorical regions

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Regions: Separate, order, align

- regions: contiguous bounded areas distinct from each other
  - separate into spatial regions: one mark per region (for now)
- use categorical or ordered attribute to separate into regions
  - no conflict with expressiveness principle for categorical attributes
- use ordered attribute to order and align regions

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Separated and aligned and ordered

- best case

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Separated and aligned but not ordered

- limitation: hard to know rank. what's 4th? what's 7th?

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Separated but not aligned or ordered

- limitation: hard to make comparisons with size (vs aligned position)

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Idiom: bar chart

- one key, one value

– data

- 1 categ attrib, 1 quant attrib

– mark: lines

– channels

- length to express quant value

- spatial regions: one per mark

– separated horizontally, aligned vertically

– ordered by quant attrib

» by label (alphabetical), by length attrib (data-driven)

– task

- compare, lookup values

– scalability

- dozens to hundreds of levels for key attrib [bars], hundreds for values

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Idiom: stacked bar chart

- one more key

– data

- 2 categ attrib, 1 quant attrib

– mark: vertical stack of line marks

- glyph: composite object, internal structure from multiple marks
  - channels
    - length and color hue
    - spatial regions: one per glyph
  - aligned: full glyph, lowest bar component
  - unaligned: other bar components
- task
  - part-to-whole relationship
- scalability: asymmetric
  - for stacked key attrib, 10-12 levels [segments]
  - for main key attrib, dozens to hundreds of levels [bars]

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Idiom: streamgraph

- generalized stacked graph
  - emphasizing horizontal continuity
- vs vertical items
  - data
    - 1 categ key attrib (movies)
    - 1 ordered key attrib (time)
    - 1 quant value attrib (counts)
  - derived data

[Stacked Graphs Geometry & Aesthetics. Byron and Wattenberg. IEEE Trans. Visualization and Computer Graphics (Proc. InfoVis 2008) 14(6): 1245–1252, (2008).]

- geometry: layers, where height encodes

counts

- 1 quant attrib (layer ordering)

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- geometry: layers, where height encodes

counts

- 1 quant attrib (layer ordering)

– scalability

- hundreds of time keys
- dozens to hundreds of movies keys

– more than stacked bars: most layers don't

extend across whole chart

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Idiom: dot / line chart

- one key, one value

– data

- 2 quant attribs

– mark: points

AND line connection marks between them

– channels

- aligned lengths to express quant value
- separated and ordered by key attrib into

horizontal regions

– task

- find trend

- connection marks emphasize ordering of items along key axis by explicitly showing relationship between one item and the next

- scalability

- hundreds of key levels, hundreds of value

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AND line connection marks between them

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Choosing bar vs line charts

- depends on type of key

attrib

- bar charts if categorical

- line charts if ordered

- do not use line charts for categorical key attribs

- violates expressiveness

principle

- implication of trend so strong that it overrides semantics!

– “The more male a person is,

the taller he/she is”

after [Bars and Lines: A Study of Graphic Communication. Zacks and Tversky. Memory and Cognition 27:6 (1999), 1073–1079.]

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Chart axes: label them!

- best practice to label
- few exceptions: individual small multiple views could share axis label

<https://xkcd.com/833/>

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Chart axes: avoid cropping y axis

- include 0 at bottom left or slope misleads

[Truncating the Y-Axis: Threat or Menace?  
Correll, Bertini, & Franconeri, CHI 2020.]

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Chart axes: avoid cropping y axis

- include 0 at bottom left or slope misleads
- some exceptions (arbitrary 0, small change matters)

[Truncating the Y-Axis: Threat or Menace?  
Correll, Bertini, & Franconeri, CHI 2020.]

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Idiom: Indexed line charts

- data: 2 quant attribs
- 1 key + 1 value
- derived data: new quant value attrib
- index
- plot instead of original value



- task: show change over time
- principle: normalized, not absolute
- scalability
- same as standard line chart

<https://public.tableau.com/profile/ben.jones#!/vizhome/CASStateRevenues/Revenues>

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Idiom: Gantt charts

- one key, two (related) values
- data
- 1 categ attrib, 2 quant attribs
- mark: line
- length: duration
- channels
- horiz position: start time

(+end from duration)

- task
- emphasize temporal overlaps & start/end

dependencies between items

- scalability
- dozens of key levels [bars]
- hundreds of value levels [durations]

<https://www.r-bloggers.com/gantt-charts-in-r-using-plotly/>

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Idiom: Slopegraphs

- two values
- data
- 2 quant value attribs
- (1 derived attrib: change

magnitude)

- mark: point + line

- line connecting mark between pts

– channels

- 2 vertical pos: express attrib

value

- (linewidth/size, color)

– task

- emphasize changes in rank/value

– scalability

- hundreds of value levels

<https://public.tableau.com/profile/ben.jones#!/vizhome/Slopegraphs/Slopegraphs>

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2 Keys

0

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Idiom: heatmap

- two keys, one value

– data

- 2 categ attribs (gene, experimental condition)

- 1 quant attrib (expression levels)

– marks: point

- separate and align in 2D matrix

– indexed by 2 categorical attributes

– channels

- color by quant attrib

– (ordered diverging colormap)

– task

- find clusters, outliers

– scalability

- 1M items, 100s of categ levels, ~10 quant attrib levels

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Heatmap reordering

<https://blogs.sas.com/content/iml/2018/05/02/reorder-variables-correlation-heat-map.html>

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Idiom: cluster heatmap

- in addition
  - derived data
- 2 cluster hierarchies
  - dendrogram
    - parent-child relationships in tree with connection line marks
    - leaves aligned so interior branch heights easy to compare
  - heatmap
    - marks (re-)ordered by cluster hierarchy traversal
    - task: assess quality of clusters found by automatic methods

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Visualization Analysis & Design

Tables (Ch 7) II

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Idioms: radial bar chart, star plot

- star plot
  - line mark, radial axes meet at central point
- radial bar chart
  - line mark, radial axes meet at central ring
  - channels: length, angle/orientation
- bar chart
  - rectilinear axes, aligned vertically

- accuracy
  - length not aligned with radial layouts
- less accurately perceived than rectilinear aligned

[Vismon: Facilitating Risk Assessment and Decision Making In Fisheries Management. Booshehrian, Möller, Peterman, and Munzner. Technical Report TR 2011-04, Simon Fraser University, School of Computing Science, 2011.]

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Idiom: radar plot

- radial line chart
  - point marks, radial layout
  - connecting line marks
- avoid unless data is cyclic

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“Radar graphs: Avoid them (99.9% of the time)”

original

difficult to interpret

redesign for  
rectilinear

<http://www.thefunctionalart.com/2012/11/radar-graphs-avoid-them-999-of-time.html>

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Idioms: pie chart, coxcomb chart

- pie chart
  - interlocking area marks with angle channel: 2D area varies
- separated & ordered radially, uniform height
  - accuracy: area less accurate than rectilinear aligned line

length

- task: part-to-whole judgements
- coxcomb chart
  - line marks with length channel: 1D length varies
- separated & ordered radially, uniform width

– direct analog to radial bar charts

- data

– 1 categ key attrib, 1 quant value attrib

[A layered grammar of graphics. Wickham. Journ. Computational and Graphical Statistics 19:1 (2010), 3–28.]

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Coxcomb / nightingale rose / polar area chart

- invented by Florence Nightingale:

Diagram of the Causes of Mortality in the Army in the East

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Coxcomb: perception

- encode: 1D length
- decode/perceive: 2D area
- nonuniform line/sector width

as length increases

– so area variation is nonlinear

wrt line mark length!

- bar chart safer: uniform

width, so area is linear with  
line mark length

– both radial & rectilinear cases

nonuniform width as length  
increases

uniform width as length increases

radial & rectilinear bars: uniform width as length  
increases

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Pie charts: perception

- some empirical evidence that  
people respond to arc length
  - decode/perceive: not angles
  - maybe also areas?...

- donut charts no worse than pie

charts

[Arcs, Angles, or Areas: Individual Data Encodings in Pie and Donut Charts.  
Skau and Kosara. Proc. EuroVis 2016.]

<https://eagereyes.org/blog/2016/an-illustrated-tour-of-the-pie-chart-study-results>

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Pie charts: best practices

- not so bad for two (or few) levels, for part-to-whole task

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Pie charts: best practices

- not so bad for two (or few) levels, for part-to-whole task
- dubious for several levels if details matter

222

Pie charts: best practices

- not so bad for two (or few) levels, for part-to-whole task
- dubious for several levels if details matter
- terrible for many levels

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Idioms: normalized stacked bar chart

- task
  - part-to-whole judgements
- normalized stacked bar chart
  - stacked bar chart, normalized to full vert height
  - single stacked bar equivalent to full pie
- high information density: requires narrow rectangle
- pie chart
  - information density: requires large circle

<http://blocks.org/mbostock/38862>

08,

<http://blocks.org/mbostock/38872>

35,

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Idiom: glyphmaps

- rectilinear good for

linear vs nonlinear trends

- radial good for cyclic patterns
- evaluating periodicity

[Glyph-maps for Visually Exploring Temporal Patterns in Climate Data and Models. Wickham, Hofmann, Wickham, and Cook. *Environmetrics* 23:5 (2012), 382–393.]

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Idiom: SPLOM

- scatterplot matrix

(SPLOM)

- rectilinear axes,

point mark

- all possible pairs of

axes

- scalability

- one dozen attribs
- dozens to hundreds of

items

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Idioms: parallel coordinates

- scatterplot limitation
- visual representation with orthogonal axes
- can show only two attributes with spatial

position channel

after [Visualization Course Figures. McGuffin, 2014.

<http://www.michaelmcguffin.com/courses/vis/>]

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Idioms: parallel coordinates

- scatterplot limitation
  - visual representation with orthogonal axes
  - can show only two attributes with spatial

position channel

- alternative: line up axes in parallel to show many attributes with position
  - item encoded with a line with  $n$  segments
  - $n$  is the number of attributes shown

- parallel coordinates

- parallel axes, jagged line for item
- rectilinear axes, item as point
- axis ordering is major challenge

– scalability

- dozens of attribs
- hundreds of items

after [Visualization Course Figures. McGuffin, 2014.

<http://www.michaelmcguffin.com/courses/vis/>]

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Task: Correlation

- scatterplot matrix
  - positive correlation
- diagonal low-to-high
  - negative correlation
- diagonal high-to-low
  - uncorrelated: spread out
- parallel coordinates
  - positive correlation
- parallel line segments
  - negative correlation
- all segments cross at halfway point
  - uncorrelated



- scattered crossings

[Hyperdimensional Data Analysis Using Parallel Coordinates. Wegman. Journ. American Statistical Association 85:411 (1990), 664–675.]

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Parallel coordinates, limitations

- visible patterns only between neighboring axis pairs
- how to pick axis order?
  - usual solution: reorderable axes, interactive exploration
  - same weakness as many other techniques
- downside of interaction: human-powered search
  - some algorithms proposed, none fully solve

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Orientation limitations

- rectilinear: scalability wrt #axes
- 2 axes best, 3 problematic, 4+

impossible

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Orientation limitations

- rectilinear: scalability wrt #axes
- 2 axes best, 3 problematic, 4+

impossible

- parallel: unfamiliarity, training

time

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Orientation limitations

- rectilinear: scalability wrt #axes
- 2 axes best, 3 problematic, 4+ impossible
- parallel: unfamiliarity, training time
- radial: perceptual limits
  - polar coordinate asymmetry

- angles lower precision than length
- nonuniform sector width/size depending on radial

distance

– frequently problematic

- but sometimes can be deliberately exploited!

– for 2 attribs of very unequal importance

[Uncovering Strengths and Weaknesses of Radial Visualizations - an Empirical Approach. Diehl, Beck and Burch. IEEE TVCG (Proc. InfoVis) 16(6):935--942, 2010.]

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Layout density

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Idiom: Dense software overviews

- data: text

– text + 1 quant attrib per line

- derived data:

– one pixel high line

– length according to original

- color line by attrib

- scalability

– 10K+ lines

[Visualization of test information to assist fault localization. Jones, Harrold, Stasko. Proc. ICSE 2002, p 467-477.]

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Arrange tables

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Visualization Analysis & Design

Network Data (Ch 9)

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